Telit EVB2.0

User Guide

1VV0301774 Rev 4
2023-11-10
Released
Confidential

Technical Documentation

Contents

1 Applicability Table

Table 1: Applicability Table

Products
All modules available on a legacy TLB
All modules available on a smart TLB

2 Introduction

2.1 Scope

This document introduces the Telit EVB 2.0. The features and solutions described in this document apply to the variants listed in the applicability table.

2.2 Audience

This document is intended for system integrators who are using the Telit module in their products.

2.3 Contact Information, Support

For technical support and general questions, e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use: https://www.telit.com/contact-us/

Product information and technical documents are accessible 24/7 on our website: https://www.telit.com

2.4 Conventions

Note: Provide advice and suggestions that may be useful when integrating the module.

Danger: This information MUST be followed, or catastrophic equipment failure or personal injury may occur.

ESD Risk: Notifies the user to take proper grounding precautions before handling the product.

Warning: Alerts the user on important steps about the module integration.

All dates are in ISO 8601 format, that is YYYY-MM-DD.

2.5 Terms and conditions

Refer to https://www.telit.com/hardware-terms-conditions/.

2.6 Disclaimer

3 General Product Description

3.1 Overview

Telit EVB 2.0 system consists of a main board, as well as connectors, cables, accessories, and software. It is designed to host Telit modules allowing users to easily test the main module functions and features.

3.2 Block Diagram

The block diagram below depicts the main Telit EVB 2.0 building blocks in their approximate positions on the board.

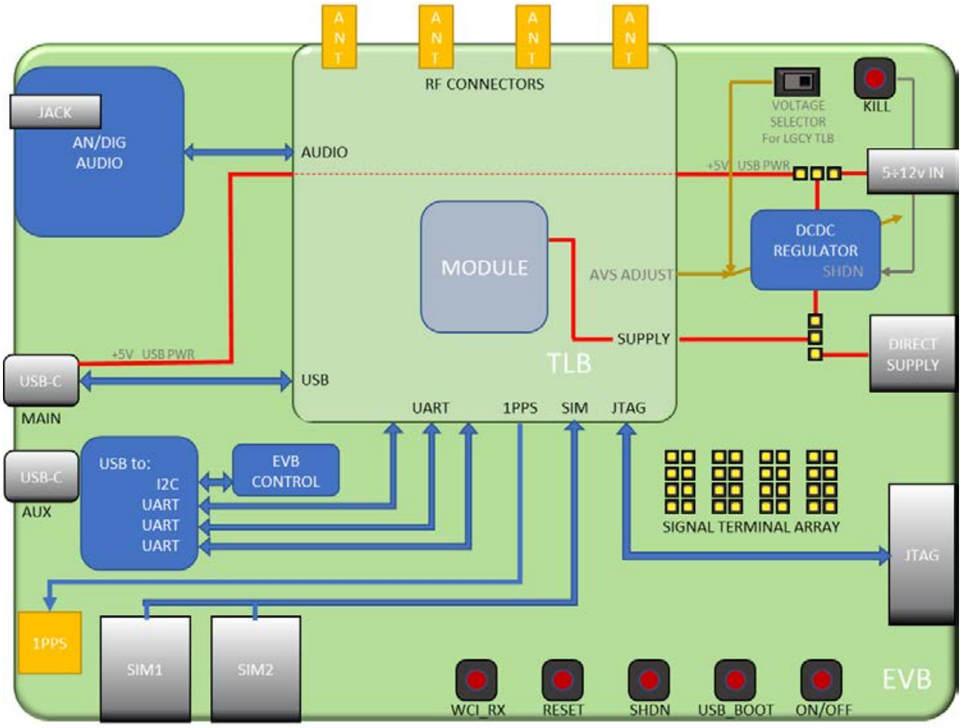


Figure 1: Telit EVB 2.0 Block Diagram

The Telit EVB 2.0 hosts a daughter board known as the TLB which serves as a mechanical adapter from various Telit modules form factors to the three board-to-board connectors.

This approach strikes a balance between the need for a small-closed-shielded hardware layout (best practice in electronics) and making it available in an open and user-friendly package. Thus, the Telit EVB 2.0 and TLBs are intentionally not optimized as a “finished” product.

Of course, some circuit parts can be used as examples or inspirations, but in some cases, end-product design solutions should be tailored to specific applications.

The modem communicates with the host PC via two USB-C cables: one for the module USB communication port and the other for serial lines, through a USB to-serial converter.

The onboard power supply on the Telit EVB 2.0 is versatile and configurable: ancillary circuits and the Telit module can be powered independently by USB PWR (DEVICE) and USB FTDI (EVB) through a regulator, via a 12V Wall Adapter through a regulator and, for specific in-depth tests, providing DC power directly from an external power supply (caution: the voltage must be regulated and needs to be within the modem voltage range).

The Telit EVB 2.0 includes two SIM card holders, thus supporting dual-SIM modems. For modules supporting only one SIM interface, the SIM1 holder is the default SIM socket.

Push buttons and LEDs are used as interfaces for the human operator.

A large number of standard 2.54 mm pin headers, clearly identified by the silkscreen on the PCB, helps the user in testing connections between ports or from an interface port to test equipment.

The audio section is minimal and allows testing with a standard hands-free headset, as well as connection to test equipment or to external amplifiers and transducers provided by the customer.

Telit modules require either 3.8, 3.3, or 1.8 V: this board is meant to be universal, thus it supports all three voltage levels; however, the user must carefully set up the board according to the instructions.

> **Warning:** In case of incorrect power supply voltage configuration, the module can be irreversibly damaged.

3.3 Telit EVB 2.0 Target

Telit EVB 2.0 design aims at full compatibility with all existing Telit products and is future-proof.

3.4 TLB Types

TLBs are classified into two types: "old" (referred to as "LEGACY TLBs") and "new" (referred to as "SMART TLBs"). They differ in whether they were designed before or after the Telit EVB 2.0 introduction.

Table 2: Product Variants and their Frequency Bands

TLB TAG	Description
LGCY38 TLB	The TLBs developed before this Telit EVB 2.0, for modules with Vbatt=3.8V. The voltage selector must be set at 3.8V.
LGCY33 TLB	The TLBs developed before this Telit EVB 2.0, for modules with Vbatt=3.3v. The voltage selector must be set to 3.3V.
SMRT38 TLB	TLBs that (despite the voltage selector position) automatically set the MAIN regulator's voltage to the required 3.8V.
SMRT33 TLB	TLBs that (despite the voltage selector position) automatically set the MAIN regulator's voltage to the required 3.3V.
SMRT18 TLB	TLBs that (despite the voltage selector position) automatically set the MAIN regulator's voltage to the required 1.8V.

Telit EVB 2.0 is backward compatible with legacy TLBs that were developed and released before its introduction.

Telit EVB 2.0 offers some special features on the smart TLB:

- Power When Needed (PWN): The Main regulator is enabled when the TLB is properly inserted.

- Automatic Voltage Selection (AVS) for main supply: the range is approximately 1-4.5V.
- Automatic Voltage Selection (AVS) for VDDIO, typically 1.8 or 2.8V.
- Automatic Voltage Selection (AVS) for VSERV. The range is roughly 0.5-3.3V.

Please refer to the "AVS Section" for detailed information about voltages on the SMART TLBs.

3.5 Main Features

Table 3: Functional Features

Function	Features
Connection with PC or external host	The USB of the device is connected to the PC via a USB-C port (DEVICE PWR) The UART ports of the device are conveyed as VCP on a USB-C port (EVB FTDI) The main UART is converted by an FTDI chip An AUX UART is converted by an FTDI chip A UART3 is converted by an FTDI chip
Versatile power supply	DIRECT SUPPLY is for advanced users (typically, hardware designers) who need to provide supply voltage directly to the module VIA REGULATOR 12V wall adapter or USB (DEVICE PWR or EVB FDTI) source selection Every device gets the proper voltage (1.8V, 3.3V, 3.8V) The voltage is automatic or even PROGRAMMABLE Power Supply can be opened for testing Power Supply is activated automatically or on demand When VIA DCDC mode is used, an external PC can control the regulator: it can be switched ON or OFF and the voltage can be set as desired
2 SIM holder	There are 2 SIM holders, for the module supporting 2 SIM interfaces SIM1 is the default SIM holder for the module that manages a single SIM.
SDIO interface	The SDIO interface goes to a dedicated connector powered by a 3.3v regulator
Digital audio	A codec provided with all ancillary parts produces the digital audio
Analog audio	Buffer amplifier, click & pop-less design without tantalum capacitors and microphone bias circuitry to manage analog audio, both connected to the jack and test points
Push buttons and switches	All the basic HW activations (On/Off, Reset, ...) can be operated manually via push buttons
LED's	LEDs provide an indication of board and module status and so on.
Current consumption measurements	Multiple ways are available to measure module current consumption.

Function	Features
	Easy disconnect of ancillary circuits (needed during normal operation) to avoid leaking currents that impact current measurements
GPIO facilitation	Module signals are tied to pin headers, to ease test and debugging. Some signals are bridged with jumpers to make them available to high-level interfaces/connectors All headers can be connected using wired jumpers
Voltage & current monitorable by PC	A current and voltage monitor connected via I2C can be read by an external PC, allowing to monitor the status of the device under test
Control extension port	The I2C controlling bus and some supply rails are available on a connector to manage a daughter board for future use
JTAG header	2 x 10 shrouded JTAG header

3.6 Main Electrical Specifications

Table 4: Electrical Specification at Connections

Connector	Specification
Direct Supply	Refer to the module specifications
5.5/2.5mm power jack	5V or 12V (36 V tolerant)
SIM holders	Refer to the specifications of the module-mounted
USB EVB FTDI	USB2 interface
USB DEVICE PWR	USB3, V_{USB} = 5v
Audio Jack Microphone	J-fet Electret Microphone: -38-45dBv/Pa
Audio Jack earpieces	>=16 Ohm, 32Ohm best

3.7 Mechanical Specifications

Dimensions

The overall dimensions of Telit EVB 2.0 are:

- Size: 156x125 [mm]
- PCB Thickness: 1.6 [mm]

Temperature Range

Table 5: Temperature Range

Mode	Temperature	Note
Operating Temperature Range	–20°C - +55°C	SIM CARDs might not withstand extreme temperatures
	–40°C - +100°C	The board (except SIM cards) is fully functional (*) across the whole temperature range.
Storage and non-operating Temperature Range	–40°C - +85°C	

Note: (*) Functional: if applicable, the board is capable to supply and enable serial/USB communications to the TLB and its module.

4 120-Pin Male B2B Connectors

4.1 B2B Connectors Layout (Top View)

The connection between Telit EVB 2.0 and TLB is implemented via three 120-poles (20 poles x 6 rows) SAMTEC SEARAY 1.27mm High Speed/High-Density B2B connectors (10mm stack height SEAM/SEAF).

The drawing below shows the B2B connectors layout on the board.

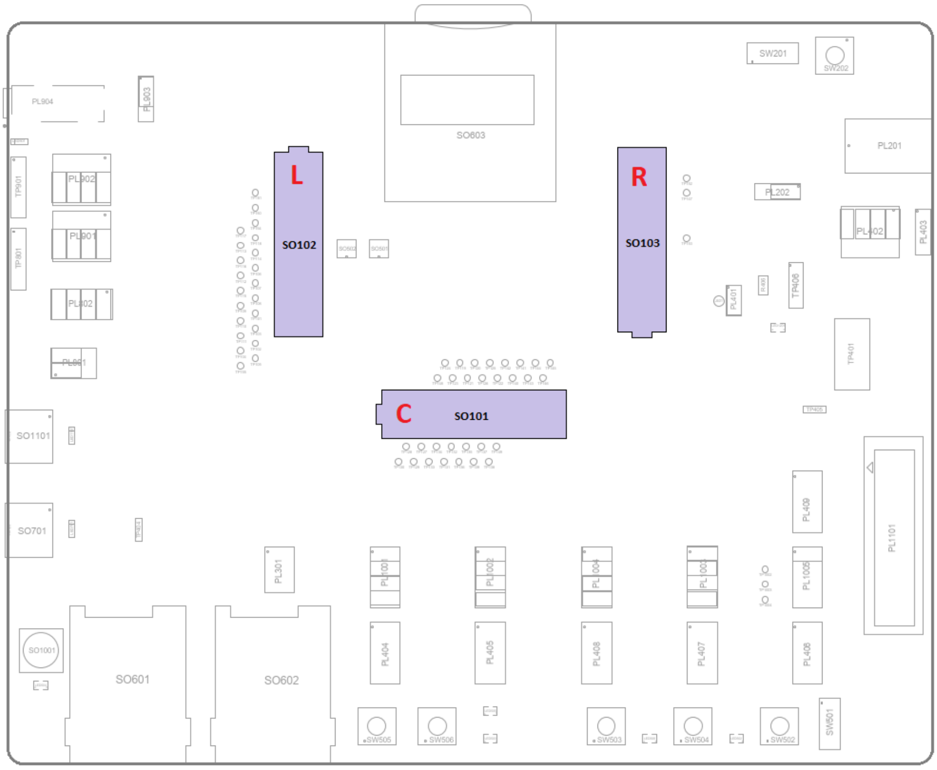


Figure 2: B2B Connectors Layout (top view)

Three tables in the following chapter provide detailed pin mapping.

Please refer to the User Guide of the specific TLB in use for the actual pin mapping and pin description.

Warning: Reserved pins must not be connected.

4.2 B2B Connectors Pinout

The tables below show the signal present on each B2B connector pin.

Table 6: B2B L Pin-out Information

B2B L (LEFT)					
1	2	3	4	5	6
GPS_LNA_BIAS	GND	GPS_LNA_EN	MICBIAS_MODULE	GND	NC / JACK_DET
7	8	9	10	11	12
GND	GND	GND	GND	GND	NC
13	14	15	16	17	18
MIC2_MT+	MIC2_MT-	GND	EAR2_MT-	EAR2_MT+	NC
19	20	21	22	23	24
GND	GND	GND	GND	GND	GND

B2B L (LEFT)					
25	26	27	28	29	30
MIC1_MT+	MIC1_MT-	GND	EAR1_MT-	EAR1_MT+	GND
31	32	33	34	35	36
SPKR_N	SPKR_P	NC	NC	RESERVED (DON'T USE)	MIC_VDD
37	38	39	40	41	42
GND	GND	D_MIC_CLK	D_MIC_DATA_1	GND	GND
43	44	45	46	47	48
NC	GND	GND	GND	GND	GND
49	50	51	52	53	54
NC	GND	GND	ADC_IN3	ADC_IN2	ADC_IN1
55	56	57	58	59	60
NC	NC	NC	NC	DAC_OUT	NC
61	62	63	64	65	66
DVI_RX	DVI_TX	DVI_CLK	DVI_WAO	REF_CLK_FF	GND
67	68	69	70	71	72
GND	GND	GND	GND	GND	ESIM_RST
73	74	75	76	77	78
GND	GND	GND	GND	SIMVCC1	SIMVCC1
79	80	81	82	83	84
HSIC_STB	HSIC_DATA	SIMCLK1	SIMIN1	SIMIO1	SIMRST1
85	86	87	88	89	90
HW_KEY	VRTC	ETH_RST_N	ETH_INT_N	SIMVCC2	SIMVCC2
91	92	93	94	95	96
USB_VBUS	USB_ID	SIMIN2	SIMIO2	SIMRST2	SIMCLK2
97	98	99	100	101	102
GND	GND	MAC_REF_CLK	MAC_TXEN_ER	MAC_MDIO	MAC_RXDV_ER
103	104	105	106	107	108
USB_D+	GND	MAC_TXD_0	MAC_MDC	MAC_RXD_0	MAC_CRS_DV
109	110	111	112	113	114
USB_D-	GND	MAC_TXD_1	MAC_TXD_2	MAC_RXD_1	MAC_RXD_2
115	116	117	118	119	120
GND	GND	MAC_TX_CLK	MAC_TXD_3	MAC_RX_CLK	MAC_RXD_3

Table 7: B2B C Pin-out Information

B2B C (CENTRAL)					
1	2	3	4	5	6
GND	GND	I2C_SCL_AUX	I2C_SDA_AUX	GND	SGMII_RX_M
7	8	9	10	11	12
USB_SS_RX_P	GND	I2C_SDA_B2B	TGPIO_06	SGMII_TX_M	SGMII_RX_P
13	14	15	16	17	18
USB_SS_RX_M	GND	TGPIO_05	I2C_SCL_B2B	SGMII_TX_P	GND
19	20	21	22	23	24

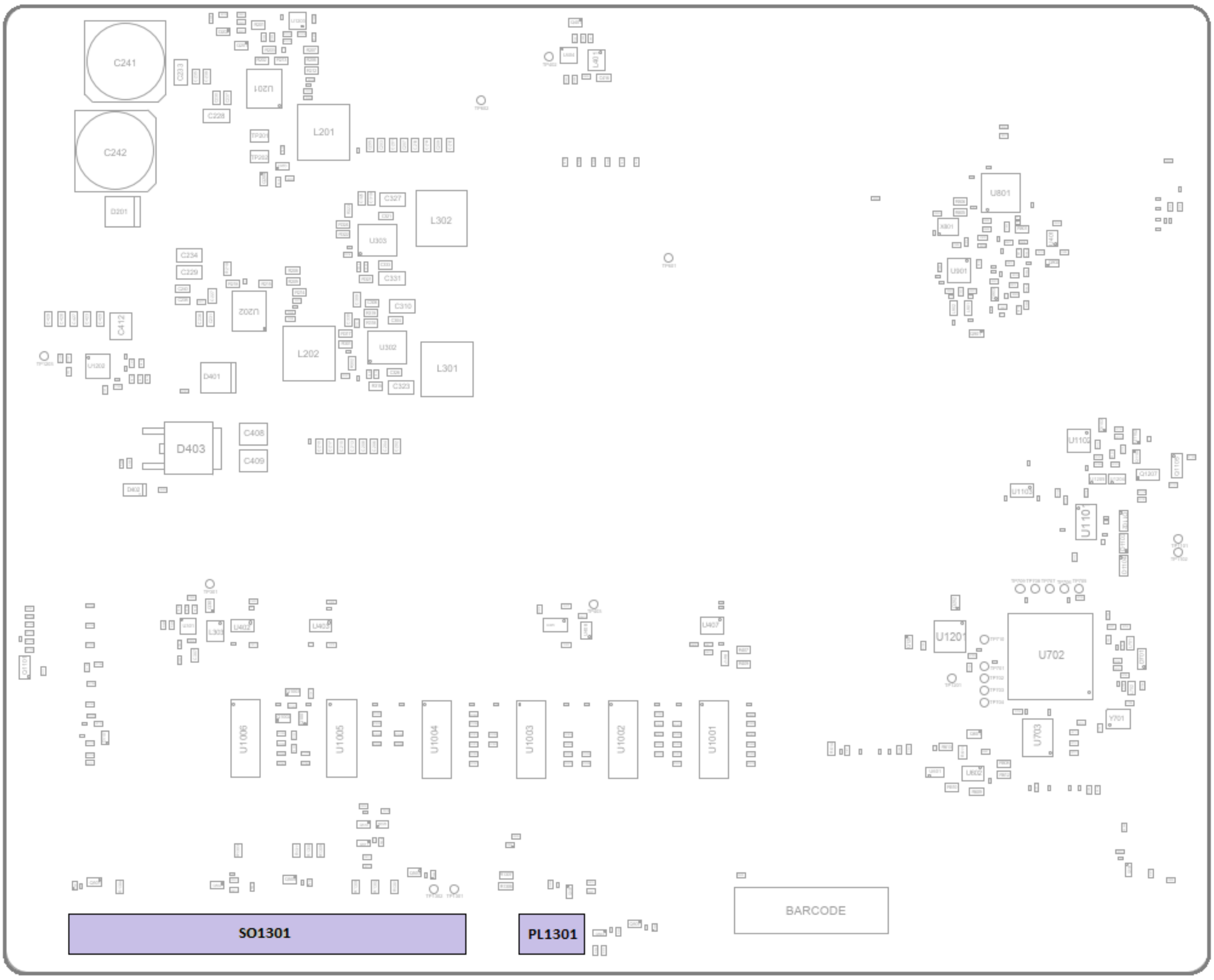
B2B C (CENTRAL)					
GND	GND	VAUX/PWRMON2	VAUX/PWRMON2	GND	PCIE_RX_P
25	26	27	28	29	30
USB_SS_TX_P	GND	NC	FORDCED_USB_BOOT	PCIE_TX_P	PCIE_RX_M
31	32	33	34	35	36
NetSO101_45	GND	TGPIO_12	SPI_MOSI	PCIE_TX_M	GND
37	38	39	40	41	42
GND	NC	TGPIO_11	TGPIO_04	GND	PCIE_REFCLK_P
43	44	45	46	47	48
#SPI_CS	TGPIO_02	TGPIO_03	SPI_MISO	NC	PCIE_REFCLK_M
49	50	51	52	53	54
VAUX/PWRMON1	VAUX/PWRMON1	LED_DRV_EN	SPI_CLK	NC	NC
55	56	57	58	59	60
TGPIO_08	TGPIO_07	TGPIO_01	TGPIO_09	NC	NC
61	62	63	64	65	66
TGPIO_21	TGPIO_10	TGPIO_22	TGPIO_20	NC	NC
67	68	69	70	71	72
VMMC	VMMC	MMC_CD	MMC_DAT3	NC	NC
73	74	75	76	77	78
MMC_DAT0	MMC_DAT2	MMC_CLK	MMC_DAT1	PCIE_EP_RESET_N	NC
79	80	81	82	83	84
TLB_CONN	GND	C107/DSR	MMC_CMD	PCIE_CLKREQ_N	NC
85	86	87	88	89	90
WIFI_SD0_TGPIO15	WIFI_SD1_TGPIO16	WIFI_SDCMD_TGPIO14	WIFI_SDRST_TGPIO13	TXD_AUX	RTS_AUX
91	92	93	94	95	96
WIFI_SD5_TGPIO24	WIFI_SD2_TGPIO17	WIFI_SD3_TGPIO18	WIFI_SD4_TGPIO23	RXD_AUX	CTS_AUX
97	98	99	100	101	102
WIFI_SD6_TGPIO25	WIFI_SD7_TGPIO26	WIFI_SDCLK_TGPIO19	WCI_TX	WCI_RX	NC
103	104	105	106	107	108
C125/RING	RFCLK2_QCA	WLAN_SLEEP_CLK	C105/RTS	PCIE_EP_WAKE_N	NC
109	110	111	112	113	114
C104/RXD	C109/DCD	C103/TXD	C106/CTS	C108/DTR	NC
115	116	117	118	119	120
NC	NC	NC	NC	NC	NC

Table 8: B2B R Pin-out Information

B2B R (RIGHT)					
1	2	3	4	5	6
VBATT	VBATT	VBATT	VBATT_PA	VBATT_PA	VBATT_PA
7	8	9	10	11	12
VBATT	VBATT	VBATT	VBATT_PA	VBATT_PA	VBATT_PA
13	14	15	16	17	18
VBATT	VBATT	VBATT	VBATT_PA	VBATT_PA	VBATT_PA
19	20	21	22	23	24
VBATT_AUX	VBATT_AUX	VBATT_AUX	VBATT_PA	VBATT_PA	VBATT_PA
25	26	27	28	29	30
GND	GND	GND	GND	GND	DCDC_ADJ
31	32	33	34	35	36
GND	LDO_ADJ	GND	DCDC_OUTPUT	3V8_EVB	VSERV_FB
37	38	39	40	41	42
NC	NC	GND	GND	GND	GND
43	44	45	46	47	48
NC	NC	NC	VSERV	VSERV	VSERV
49	50	51	52	53	54
NC	NC	NC	NC	NC	NC
55	56	57	58	59	60
NC	NC	UART3_TXD	UART3_RXD	UART3_RTS	UART3_CTS
61	62	63	64	65	66
NC	NC	NC	NC	NC	TP153
67	68	69	70	71	72
NC	NC	NC	NC	NC	NC
73	74	75	76	77	78
NC	NC	NC	NC	NC	JTAG_DETECT
79	80	81	82	83	84
OLD_TLB_GND	GND	GND	TLB_DETECT	GND	GND
85	86	87	88	89	90
GND	GND	GND	GND	GND	GND
91	92	93	94	95	96
#RESET	#ON_OFF	STAT_LED	LED_DRV	SW_RDY/SYSTEM_ON	#SHDN
97	98	99	100	101	102
GND	GND	GND	GND	JTAG_TRIGOUT	JTAG_TRIGIN
103	104	105	106	107	108
GPS_PPS	GPS_RFPAON	GPS_CLK	GND	JTAG_SUPPLY	JTAG_PS_HOLD
109	110	111	112	113	114
GND	GND	GND	GND	JTAG_TDI	TP152
115	116	117	118	119	120
JTAG_TMS	JTAG_TDO	#JTAG_TRST	JTAG_TCK	JTAG_RTCK	#JTAG_RESOUT

5 Extra Connectors on Bottom Side

5.1 Extra Connectors Layout (Bottom View)



SO1301 2x20 Female

Reserved for future use.

PL1301 2x3 Female

Reserved for future use.

6 Power Supply

During the final product development phases, it might be necessary to supply the module with low-level voltages (either 1.8V, 3.3V, or 3.8V) from an external DC source located close to the board and through a wire of adequate section. This case is defined as "DIRECT SUPPLY".

Other users will just need to start working with the target module quickly and with reduced external devices. In this use case, a wall adapter connected to the board through the on-board regulator will be the optimal solution. This case is defined as "VIA DCDC".

Both approaches are supported with Telit EVB 2.0, hybrid solution is also possible.

Quite often, a user must source the target in DIRECT SUPPLY mode (i.e., to be closer to a "real application" situation, or to observe the module current consumption) while also powering up the remaining EVB circuitry with a separate power supply (that is, via DCDC).

For each of the four rails, the following options are available:

- V_{batt}, feeding the baseband part of the target module
- V_{batt}_PA, feeding the RF part of the target module
- V_{batt}_AUX, feeding the auxiliary circuits NOT belonging to the module
- 3V8_EVB, feeding auxiliary Telit EVB 2.0 circuits, fixed at 3.8V

These voltages are displayed on the selector from left to right, as shown below.

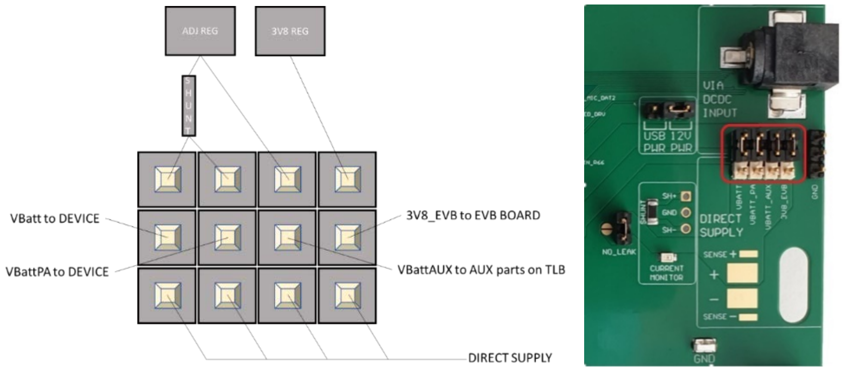


Figure 3: 4 Ways Power Selector

> **Note:** VIA DCDC can in turn be sourced by the wall adapter or, using the 3x1 male header selector, by the 5v of the USB DEVICE PWR.

6.1 Example Use Cases

Use Case 1:

The TLB is powered externally by DIRECT SUPPLY, while the Telit EVB 2.0 is powered by VIA DCDC. The Telit EVB 2.0 must always receive 3.8 V, so it is most frequently connected to the internal fixed regulator.

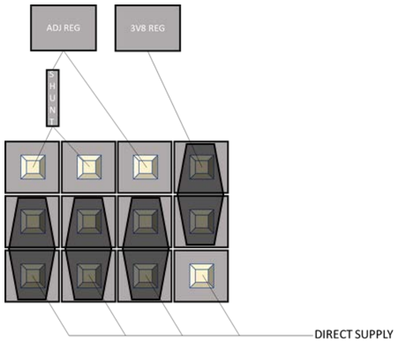


Figure 4: Use Case 1: 4 Ways Selector

Warning: This rail can only be connected to DIRECT SUPPLY when the external power supply is set to 3.8V.

Use Case 2:

The two most central headers on the left are the device's supply terminals, so if a separate DC source is required, connect to these two headers.

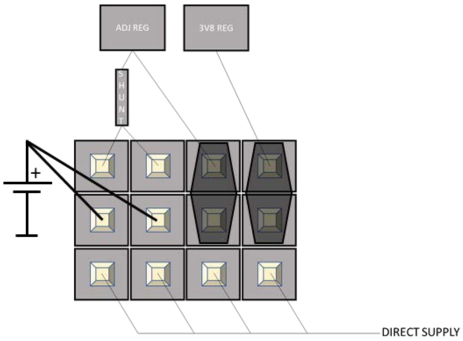


Figure 5: Use Case 2: 4 Ways Selector

Note: Refer to section 5.6 Power Consumption Measurement for more information and use cases.

6.2 V_{batt} Voltage Level

Telit modules typically require either 3.8V, data cards require 3.3V, and 1.8V is the standard voltage for positioning modules. Please follow the guidelines below to provide the correct voltage to each Telit module:

When in DIRECT SUPPLY mode,

The user is responsible for setting the correct voltage on the external power supply.

When in VIA DCDC mode,

For LEGACY TLB’s

The voltage for LEGACY TLBs is selected using the EVB's 3.3v/3.8V voltage selector (please remember no LEGACY TLB supports positioning modules, thus there is no need for 1.8V selection)

For SMART TLB

The voltage is automatically set (that is AVS, Automatic Voltage Selection is implemented). A resistor mounted on the TLB properly biases the voltage regulator network and triggers the V_{batt} voltage to be set according to the module mounted on the TLB.

6.3 Power Supply Configuration

No-Leak Jumper

NO_LEAK is a jumper that disconnects reverse polarity protection diodes, over-voltage protection, and stabilization capacitors. These components introduce current leakage thus this jumper must be removed during current consumption tests.

The adjacent "coffee bean" is a solder bridge that can be easily bridged with a solder ball and is placed parallel to the no-leak jumper.

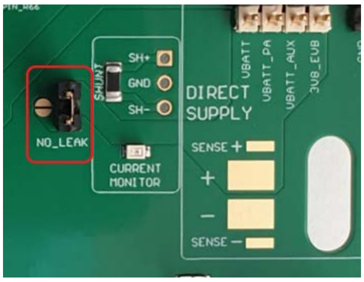


Figure 6: No-leak Jumper

Source Selector

When the VIA DCDC mode for some supply rail is used, it is possible to select the power source that can be supplied.

Power source from the wall adapter (5V-12V, connected to the 5.5x2.5 mm jack): when the jumper is moved to the right position.

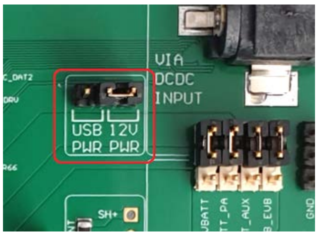


Figure 7: Source Selector in 12V position

Power source from the 5V USB-PWR (USB DEVICE PWR): when the jumper is moved to the left position.

Figure 8: Source Selector in USB-PWR position

Power source from the 5V USB-FTDI (USB EVB FTDI): when the central pin is directly connected via a wired jumper to the pin "USB_EVB" of the "DOMAINS" connector (look at the chapter Voltage Domains for further details about this connector).

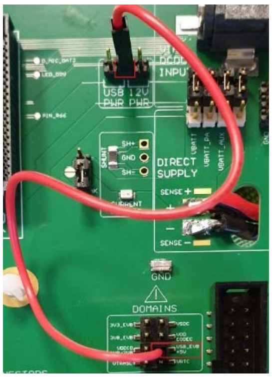


Figure 9: Source from USB-FTDI

Voltage Selector

On the legacy TLB board (developed before Telit EVB 2.0 introduction), no AVS (Automatic Voltage Selector) is available, thus the DCDC regulator must be set to the voltage needed by the device using the voltage selector switch.

- With the selector set on the left position, the voltage is set to 3.3V.
- With the selector set in the right position, the voltage is set to 3.8V.

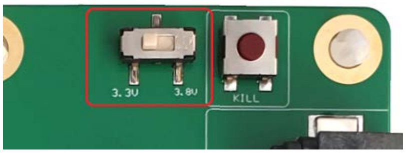


Figure 10: Voltage Selector

Note: On the "SMART TLBs" this selector does not affect voltage selection.

Kill Push Button

The KILL button, located in the top right-hand corner of the Telit EVB 2.0, is designed to choke the device by disabling the DCDC regulator.

It is not recommended to use this button to power down the Telit module because the device must be turned off properly. This feature is only available to allow the user to simulate a sudden power supply interruption.

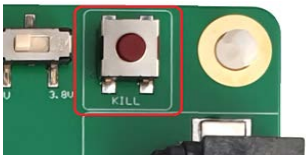


Figure 11: Kill Push Button

6.4 Power Supply Requirements

In DIRECT SUPPLY mode, the wirings impedance should be kept under control and must be less than 150mΩ, with an inductance not greater than 1µH. In general, a 0.5m long 18AWG wire is suitable.

In VIA DCDC mode, the wall adapter connected to the jack can supply a voltage in the 5V-12V range. The regulator can accept up to 36V (42V absolute maximum) but, in this case, the output voltage (V_{batt}) will not be properly regulated.

Note: A minimum current of 1.25A is required.

The cylindric jack is a 5,5x2,5 mm type with the inner terminal positive.

The WALL ADAPTER must fulfill the following requirements:

Table 9: WALL ADAPTER Requirements

Wall Adapter	Value
Nominal Supply Voltage	8 V ÷ 12 V
Operating Voltage Range	4.8 V - 13 V (TBC)
Extended Voltage Range	4.8 V - 24 V (TBC)
Current sourced	>=1.25A

Warning: When the power supply wire voltage drops to 4.8-5V (especially with thin wires), the 3.8V module may experience undervoltage.

6.5 4-Wires Connection for Direct Supply

Any voltage drop due to connections or high-impedance wires is a potential issue for the target and, in some cases, can lead to unexpected device switch-off (when V_{batt} falls below the minimum threshold). Thus, to prevent issues, it is recommended to use short and thick wires to supply power to the Telit EVB 2.0.

If possible, use the 4-wire connection (2 source cables and 2 sense cables), connecting all wires at the endpoint on the four dedicated pads on the Telit EVB 2.0. Termination" capacitors - requested by the external Power Supply to be stable – are already mounted on the Telit EVB 2.0.

A long cable without detection could be affected not only by voltage drops, but could also lead to power supply instability (that is voltage overshoot, thus permanent device failure).

Direct Supply

Soldering is the best connection method to prevent voltage drops. To ensure good mechanical retention as well, it is recommended to pass the wires through the cable lock slot.

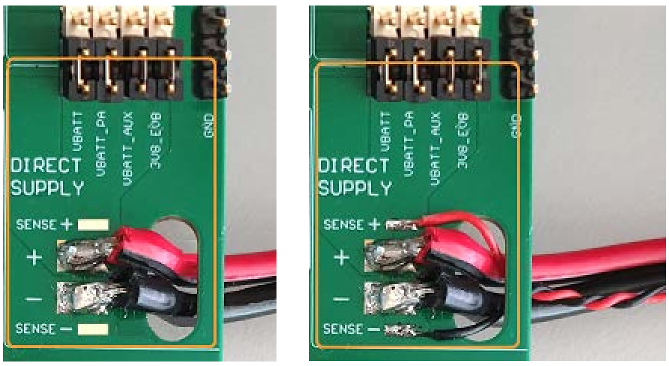


Figure 12: Direct Supply and Direct Supply with Sense

6.6 Power Consumption Measurement

The Telit EVB 2.0 not only allows different power supply configurations (as explained in the previous chapter), but at the same time supports different methods to measure the current consumption of the Telit module connected to the Telit EVB 2.0 :

Current measurements through a current meter (EVB powered VIA "4 Ways Selector")

Current measurements current through an external power supply (Telit EVB 2.0 powered via "Direct Supply")

Current measurements through a shunt resistor.

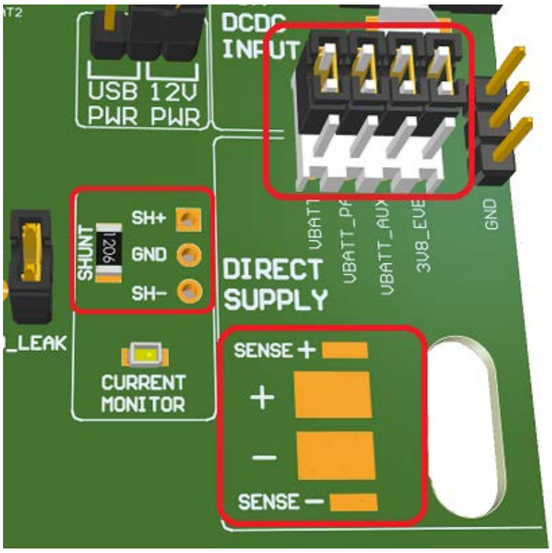


Figure 13: Current measurement points: 4 Ways Selector, Direct Supply, and shunt resistor

With the first two methods, it is possible to measure the current separately for **VBATT**, **VBATT_PA**, **VBATT_AUX**, and **3V8_EVB** rails.

> **Note:** Each current component can be measured separately only if allowed by the TLB geometry. Please check the TLB schematic to verify if these paths are available separately.

Measuring the Current by a Current Meter

The board is fully powered VIA DCDC. The current consumption of the device can be measured by placing an Ammeter in series to the desired rail.

The image below shows the VBATT measurement configuration.

Please ensure that the ammeter wires are properly sized. Avoid using thin or long wires because their high impedance could lead to voltage drop and the device could malfunction (disconnection, switch off).

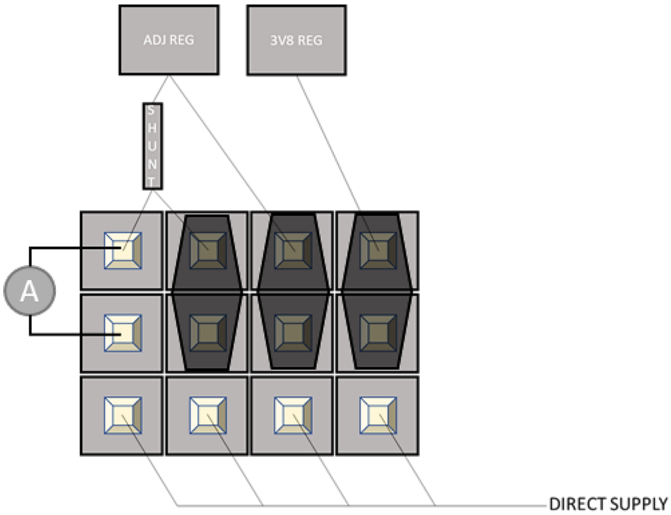


Figure 14: Ammeter Insertion

Measurement setup example:

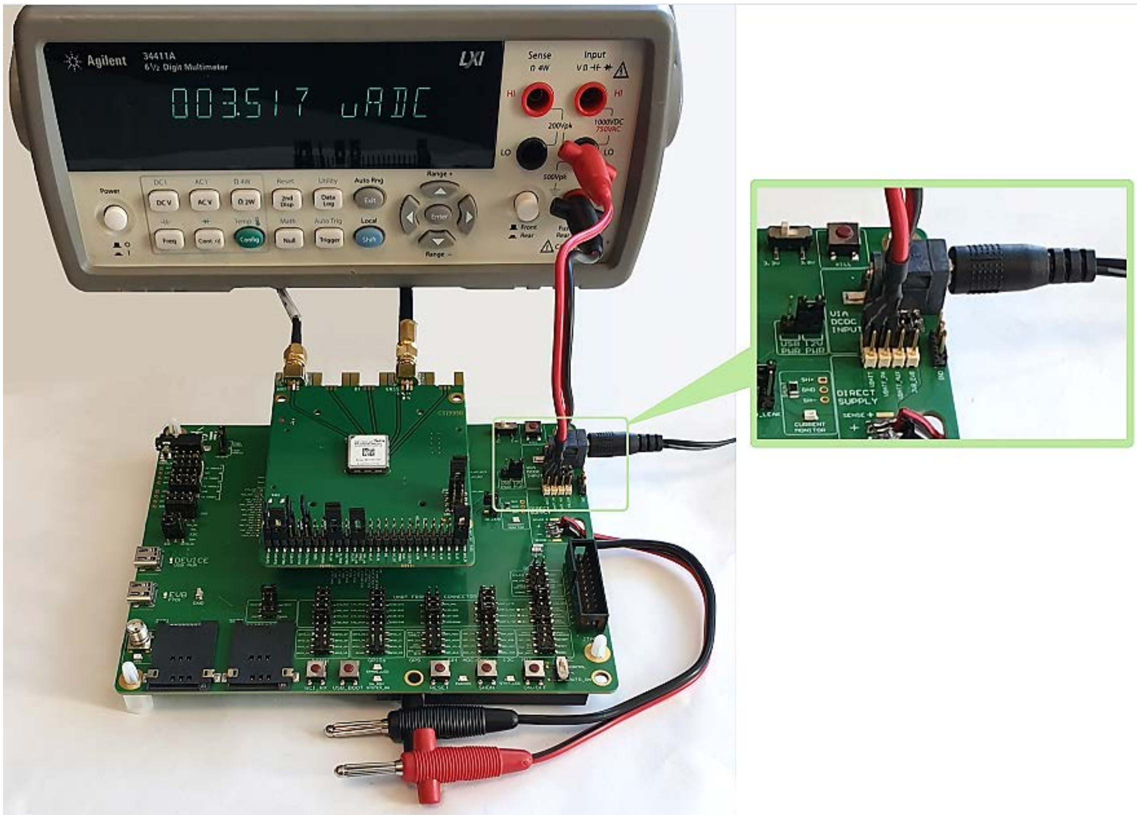


Figure 15: Measurement Setup with Ammeter

Measuring the Current by a Power Supply

In this case, only the device is powered by an external source via DIRECT SUPPLY, while the TLB's auxiliary parts and the Telit EVB 2.0 board are powered via DCDC. A key requirement is that the external power supply must be able to perform current measurements.

The image below shows how to measure the consumption of the device.

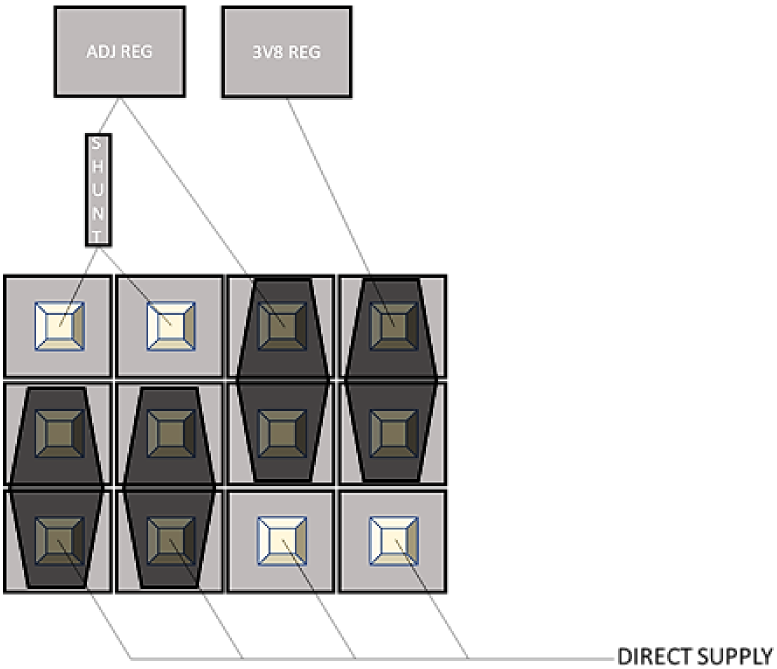


Figure 16: Measure the Current by a Power Supply

Example of measurement setup:

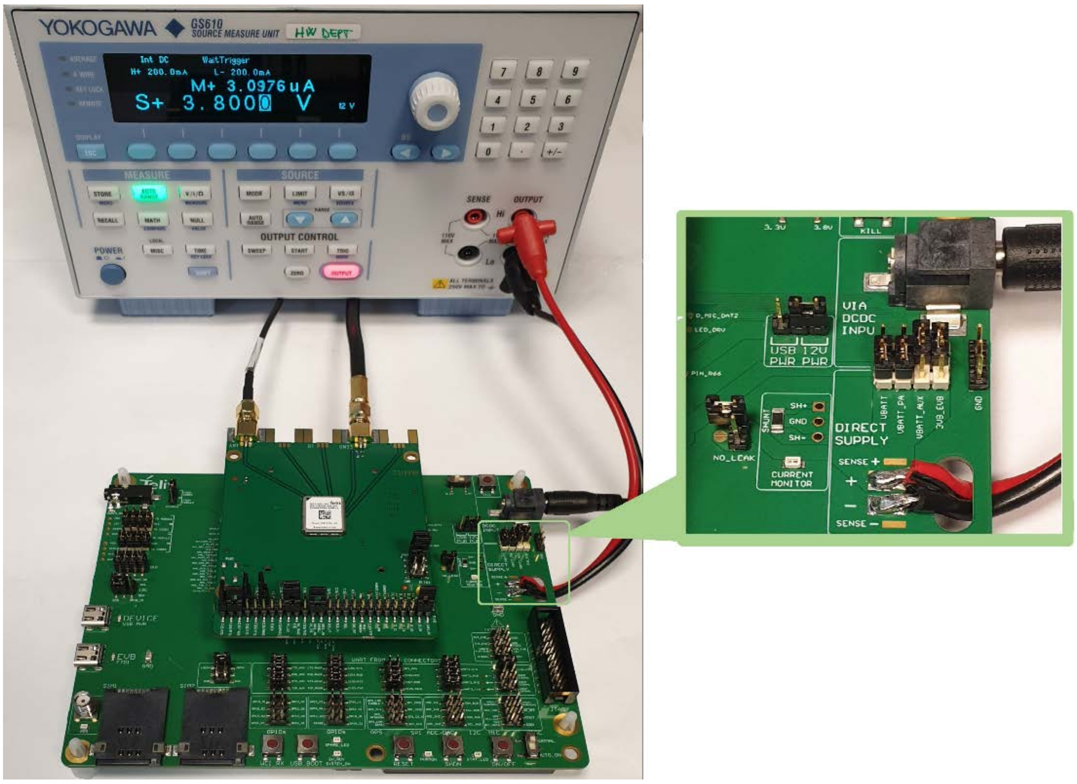


Figure 17: Measurement Setup with Ammeter

Note: Remove the NO-LEAK jumper to avoid leakage currents due to the protection network and the stabilization capacitors.

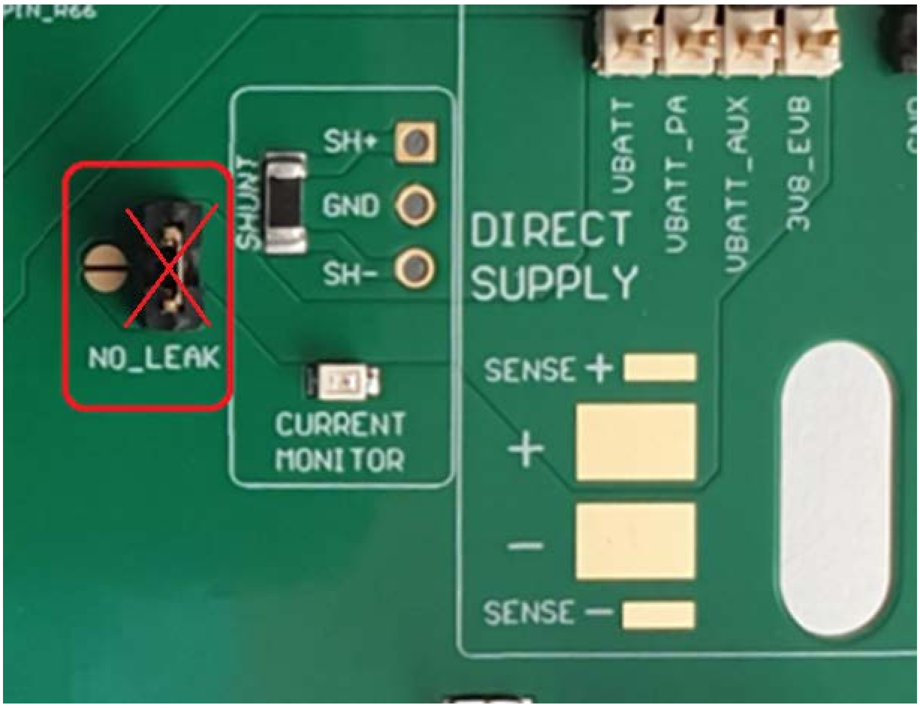


Figure 18: Avoiding Leakage Current

Measuring the Current through the Shunt Resistor

The device is exclusively powered via DCDC.

The device current consumption (total current consumption on VBATT and VBATT_PA) is indirectly measured by measuring the voltage across the 50 mΩ shunt resistor.

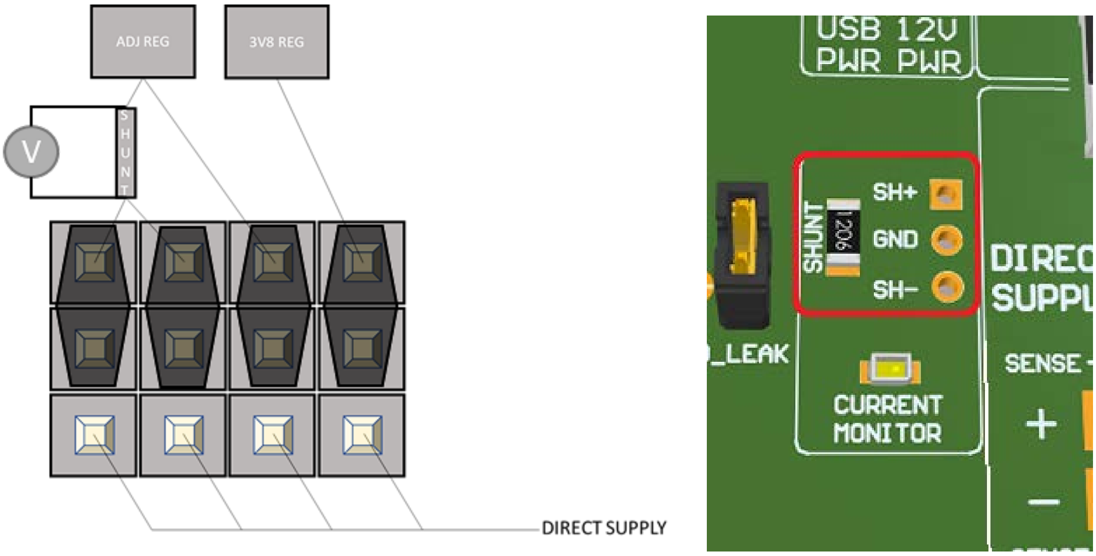


Figure 19: Measure the Current through the Shunt Resistor

6.7 VAUX/PWRMON Power Output

Most Telit modules include a regulated power supply output to supply small devices from the module itself, such as level translators, audio codecs, sensors, and so on.

Due to the different architectures, devices provide different voltages (usually 1.8V, rarely 2.8V) and current values, some as high as 50mA, some as low as 1mA. The current flowing through this port adds up to the Vbatt current consumption. This is fine when the module is mounted on the final application, but is not acceptable when it is on top of the Telit EVB 2.0, because in this way the consumption measured on the Telit EVB 2.0 is not in line with the specs (recall that the specs can not know in advance the currents drained from the ports of the module). So, on the Telit EVB 2.0, this Vaux/powermon port is buffered.

On the final application, the buffer is needed only if the load current exceeds the Vaux/powermon current budget.

At the same time, if desired, VAUX/PWRMON1/2 pins are available for supplying the translators through a 0-Ohm jumper. They are available as well on their pin header to allow the user to connect external application hardware and test the system.

> **Note:** When connecting a load on this net, the related current will bias positively the current drawn by the device. Please take this into account when evaluating device power consumption.

7 VOLTAGE DOMAINS

All voltage domains are grouped in the frontal area of the board and are available on a dedicated male connector (2x4 pins, 2.54 mm): on this connector, each pin represents a specific voltage domain. Each voltage name is silk-screened on the board to allow quick identification.

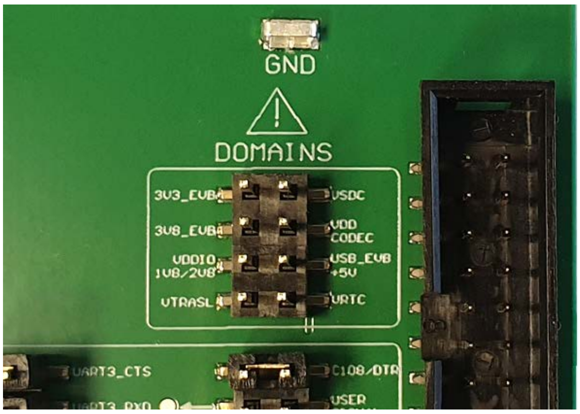


Figure 20: Voltage Domains

This connector makes testing and debugging operations easy because it allows internal or external connections using wire jumpers.

> **Warning:** Use this connector only for voltage monitoring or to power light loads: except for "USB_EVB +5V" do not exceed 5mA.
>
> In case of overload, short circuit to ground, or shorts between different voltage domains, the equipment will be irreversibly damaged
>
> Don't connect an external power supply to these pins or catastrophic equipment failure may occur.

> **Note:** Since the +5v VBUS of the "EVB FTDI" USB connector is brought to the Pin "USB_EVB +5V" of the DOMAINS connector, using a wired jumper to connect it to the central pin of "Source Selector", is it possible to take the +5v supply to the input of the DCDC regulators. This allows to supply of the Telit EVB 2.0 using the USB connected to the FTDI chip. This connection is useful but could be not so ideal (because of the voltage drops on the wired jumper) and has to be tested in case of high current consumption (i.e. module in GPRS connection with 4 active TX-slots).

8 Digital Section

Most of the device pins available on the Telit EVB 2.0 are digital.

Its supply domain is VDDIO_1V8/2V8 and is set to 1.8V or 2.8V depending on the device (legacy TLBs are satisfied with 1.8V, smart TLBs set the voltage on its own).

These signals mostly terminate on a male connector, allowing the user to connect to other boards using wire jumpers.

Some of these lines are not terminated on a header and are connected directly to a peripheral (that is the serial or the DVI lines). In such cases, there are a couple of headers with jumpers: this allows the signals to function normally, but also to be interrupted to facilitate testing, debugging, and measuring.

Where possible, the name of the signal is silk-screened on the board to allow quick identification.

Digital pins are grouped in the frontal area of the board and are available on 2x4 pins, and 2.54 mm connectors.

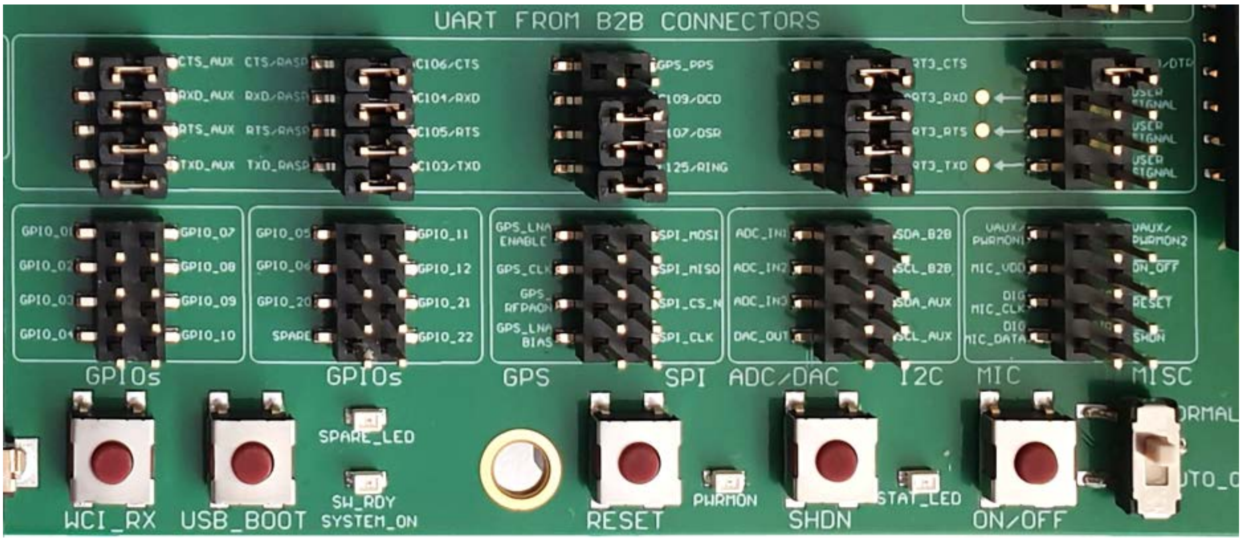


Figure 21: Digital Section: Pinout

8.1 Power On

In most cases, a button connecting a pin to the ground is the proper way to turn on Telit devices.

In the other cases, there will be an adapter circuitry on the TLB that translates the open collector interface with the appropriate one, depending on the specific device.

> **Note:** In this document, all inverted lines (that is, active low signals), are labeled with a name ending with '#','*', or with a bar above the name.
>
> **Warning:** To check if the device has powered on, the hardware line PWRMON should be monitored.
> No pull-up resistor should ever be used on the ON_OFF* line since it is internally pulled up. Using a pull-up resistor may cause problems with improper latching and power on/off of the module. The ON_OFF* line must be connected only in an open collector or open drain configuration.

8.2 Communication Ports

USB DEVICE (USB PWR) and USB EVB (FTDI)

The Telit EVB 2.0 mounts two USB-C connectors labeled "DEVICE" and "EVB".

Normally, communication between the TLB-mounted device and the host PC takes place either through a USB DEVICE port or UART lines (translated by the FTDI chip).

Figure 22: USB DEVICE (USB PWR) and USB EVB (FTDI)

The USB DEVICE is the upper one and is connected to the Telit module USB port. Through this port it is possible to power the Telit EVB 2.0, provided that the source selector is set on the left position (refer to 0 Source Selector).

> **Note:** Some TLBs provide an onboard USB connector instead of relying on the USB port mounted on the EVB. Please refer to the TLB hardware documentation for further information.

The USB EVB (FTDI) is connected to an FTDI 4-channel port translator splitting the USB bus into one I2C interface and three UART lines.

The 3 UART lines are connected to the device serial lines and are mapped as Virtual Com Ports on the host PC.

> **Note:** When the USB DEVICE (USB PWR) is used as a power source, make sure the USB cable is as short and thick as possible, to avoid voltage drops on the cable that can cause issues.
>
> Please note that a regulator must have some voltage headroom to produce a reliable output voltage and the voltage drops reduce this margin.

UARTs Pins Naming Convention

On the Telit EVB 2.0 schematic, there is an apparent misalignment between the pin's direction and its naming.

> **Note:** Pins direction is inverted because, on the main UART, the host is considered the DTE (as per V.24 standards). On the other hand, for the other 2 UART lines, the Telit device is rated as DTE so the host becomes a DCE.

Serial Ports

The three serial ports are listed below:

- MODEM SERIAL (Main) is the port dedicated to the AT commands
- AUX SERIAL is a second port for modules supporting two serial ports
- UART3 SERIAL is a spare serial port, often used by the GNSS receiver integrated into cellular devices

Each serial line must be configured as 115200 8-N-1 unless otherwise specified.

Serial port lines can be monitored (for example with a scope or logic state analyzer) or disconnected, thanks to dedicated jumpers placed on the front of the board. Each line jumper is silk-screened to allow easy identification. These lines operate at the same voltage as the module under test, as they are the device communication ports.

This allows to interface of the Telit device with the customer hardware prototype.

Note: For minimal implementation, only TXD and RXD lines can be connected, with the other lines left floating as long as software flow control is implemented.

To avoid back powering effect, it is recommended to prevent HIGH logic level signals from being applied to the digital pins when the device is powered off or during an ON/OFF transition.

8.3 SIM Card Holders

Since some cellular modules support two SIM interfaces, the Telit EVB 2.0 mounts two SIM card holders.

The default SIM holder is the one labeled“SIM1” on the left.

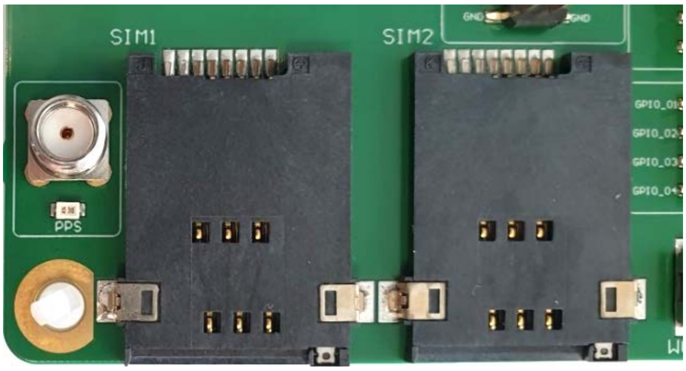


Figure 23: Sim Holders

Warning: Troubleshooting in SIMIN detection: for EMC reasons, a capacitor has been placed on the SIMN1 and SIMIN2 switching lines, which, for some modules, may be excessive and indicate "SIM present" when using the AT command "CPIN?" for SIM detection even though it is not there: in this case, the solution is to remove, from the EVB 2.0, these capacitors (see the image below).

On your application: leave these capacitors not populated or, if necessary, mount capacitors on SIMN1 and SIMIN2 lines if suggested in the hardware user guide for the specific module.

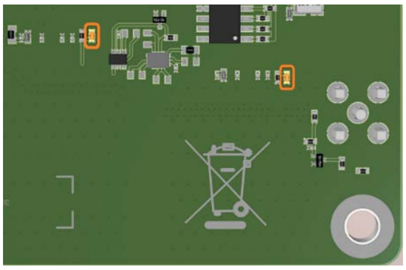


Figure 24: Bottom side view: highlighted the capacitors that must be removed in case of SIM detection error

8.4 SDIO Interface

At the top center of the Telit EVB 2.0, an SDIO card connector is mounted.

SDIO cards are supplied with 3.3V: this rail (derived from the 3V8_EVB line) is always on when the board is powered on.

Figure 25: SDIO Interface

8.5 LEDs

Current Monitor

When exceeding about 100mA current draw, the LED switches on.

Figure 26: Current Monitor LED

STAT_ LED

Indication of Network Service Availability

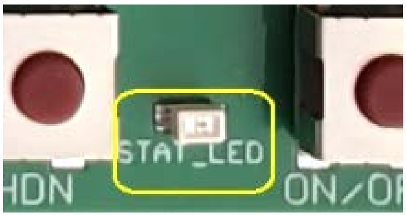


Figure 27: STAT LED

The STAT_LED pin status displays network service availability and call status. The function is available as an alternative function of GPIO_01 (to be enabled using the AT#GPIO=1,0,2 AT command). Please refer to each module documentation for information on network service LED status.

STAT_LED status is defined according to the table below:

Table 10: LED Status

Device Status	Led Status
Device off	Permanently off
Not Registered	Permanently on
Registered in idle	Blinking (1s on + 2 s off)
Registered in idle + power saving	Depends on the event that triggers the wakeup (In sync with network paging)
Connecting	Blinking 1s on + 2s off

PWRMON

This LED switches on when the module is operative.

Figure 28: PWRMON LED

SW_RDY/SYSTEM_RDY

This LED switches on when the module is ready to operate.

Note: Only some Telit devices support this function.

Figure 29: SW_RDY/SYSTEM_RDY LED

SPARE_LED

This LED is a visual test probe that switches on when the test point (a pin labeled "SPARE" located in the header connector placed near the LED) connected by the user reaches the high logic level.

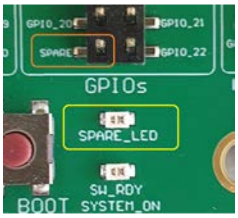


Figure 30: SPARE LED and

GPS PPS

This LED switches on when the device enables 1PPS output and the level is high: it's the 1PPS display, typical for positioning devices.

To make this signal (level 3.3V) available for external use, an SMA connector has been placed near the homonymous led

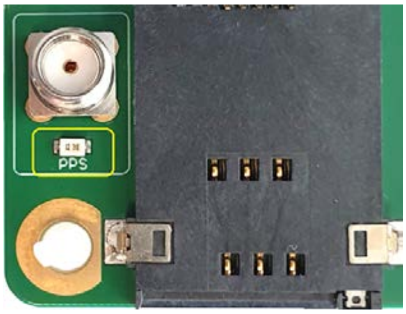


Figure 31: PPS LED

USB DEVICE ON

This LED switches on when the USB DEVICE is powered, and the USB controller is enabled.

Figure 32: USB DEVICE LED

USB EVB FTDI ON

This LED switches on when the host PC operating system enumerates the USB instance of the FTDI-level converter.

Figure 33: FDTI LED

Audio ON

This LED switches on when the “carrier” signal is present on the audio jack. It detects when the balanced output lines reach ½ VDD stage and is ready to produce sound.

The indication is supported for both digital and analog audio chains.

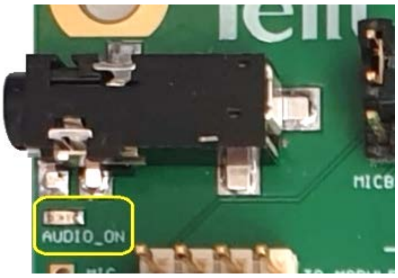


Figure 34: AUDIO_ON LED

8.6 Push Buttons

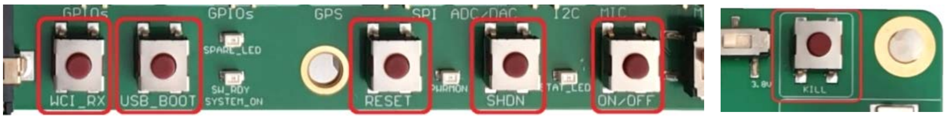


Figure 35: Push buttons

> **Warning:** Troubleshooting in push buttons functions: for EMC reasons, capacitors have been placed on the USB_BOOT, RESET, SHDN, and ON/OFF lines, which, for some modules, may be excessive and could be the cause of unwanted behavior. For example, the module may go in BOOT-Loader mode as USB_BOOT would be pressed.
>
> To avoid unwanted behavior, it is strongly recommended to remove from the bottom side of the EVB 2.0, the following four capacitors:

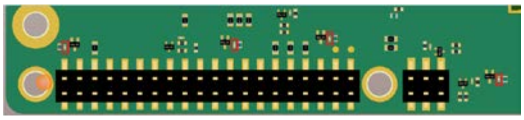


Figure 36: Troubleshooting buttons

WCI_RX

This push button is available for compliance with some existing Telit devices requiring this line.

USB_BOOT

FORCED_USB_BOOT pin must be activated only during the firmware upgrade operation. Normally it must be left idle.

RESET

Some devices require a low level on this line as a RESET command.

SHDN

Some devices require a low level on this line as a SHUTDOWN request.

ON/OFF

This is the ON/OFF push button: keeping it pressed for a few seconds switches the device ON or OFF.

The adjacent “AUTO_ON” switch, is a “comfort gadget” for users who want to emulate the ON/OFF button always pressed. When activated, it “holds down” the ON/OFF push button permanently.

KILL

This push button abruptly stops the main DCDC regulator, causing a power disruption if a device is powered through DCDC. It is the forced power-cycle function.

> **Warning:** Use of KILL is not recommended.
>
> Make sure to follow the switch-off procedures for the specific device in the application. It is provided on the Telit EVB 2.0 for testing purposes only.

9 Audio Section

The Telit EVB 2.0 AUDIO interface includes a headphone amplifier, typically not present on Telit evaluation boards: this solution was chosen since loudspeakers are referenced to ground, thus avoiding tantalum DC block capacitors causing audio clicks and pops.

The user interface is a CTIA standard hand-free TRRS 3.5mm jack or a 4 test points array in 2.54mm pitch.

Figure 37: Digital Audio Jumpers Configuration

Since a CODEC (provided with a local clock and I2C pull-up) handles the conversion, the other end, towards the device under test, can be easily operated as an analog or digital interface.

This I2C interface (necessary to configure and activate the codec) is different from that of the FTDI chip: it belongs to the I2C port of the module under test.

Some Telit modules have dedicated I2C pins, known as "native I2C", while others share lines between GPIO and I2C.

The I2C selector allows one to choose between native I2C and GPIO1/2: if different GPIO lines must be used, it is suggested to connect them using the supplied wired jumpers.

To provide the user with maximum flexibility, all audio and codec signals are routed through a pair of jumpered headers. For example, a handsfree can be connected directly and jumpers can be removed to connect external circuitry.

To initialize the audio codec using GPIO1 as SDA and GPIO2 as SCL, the following commands from the AT interface must be sent:

```
AT#I2CWR=1,2,30,4,19
00109000100A330000330C0C09092424400060 <CTRL-Z>
AT#I2CWR=1,2,30,17,1
8A <CTRL-Z>
```

Note: These setup commands via I2C have to be re-issued after the Module gets back from sleep.

Warning: ATroubleshooting at modules startup: if a module gets stuck at startup, please check the jumpers setting for the audio codec: if connected to GPIO1/2, remove the jumpers or keep them in the left “B2B” position on the I2C selector (see Figure 38) until the module boots up.

To use the audio in ANALOG MODE, the jumpers must be set as shown in the image below:

Figure 38: Jumpers Configuration for Audio in Analog Mode

Telit module's digital audio interface (DVI) is based on the I2S serial bus interface standard. The audio port can be connected to the end device using the digital interface, or via one of the several compliant codecs (in case an analog audio is needed).

9.1 Electrical Characteristics

The product is providing the DVI on the following pins:

Table 11: DVI Pins

Signal	I/O for the device	Function	Type
DVI_WA0	Output for Device (MASTER) Input for Device (SLAVE)	Digital Audio Interface (Word Alignment / LRCLK)	CMOS 1.8V/2.8V
DVI_RX	Input for Device Output for Codec	Digital Audio Interface (RX)	CMOS 1.8V/2.8V
DVI_TX	Output for Device Input for Codec	Digital Audio Interface (TX)	CMOS 1.8V/2.8V
DVI_CLK	Output for Device (MASTER) Input for Device (SLAVE)	Digital Audio Interface (BCLK)	CMOS 1.8V/2.8V

10 Mechanical Design

10.1 Drawing

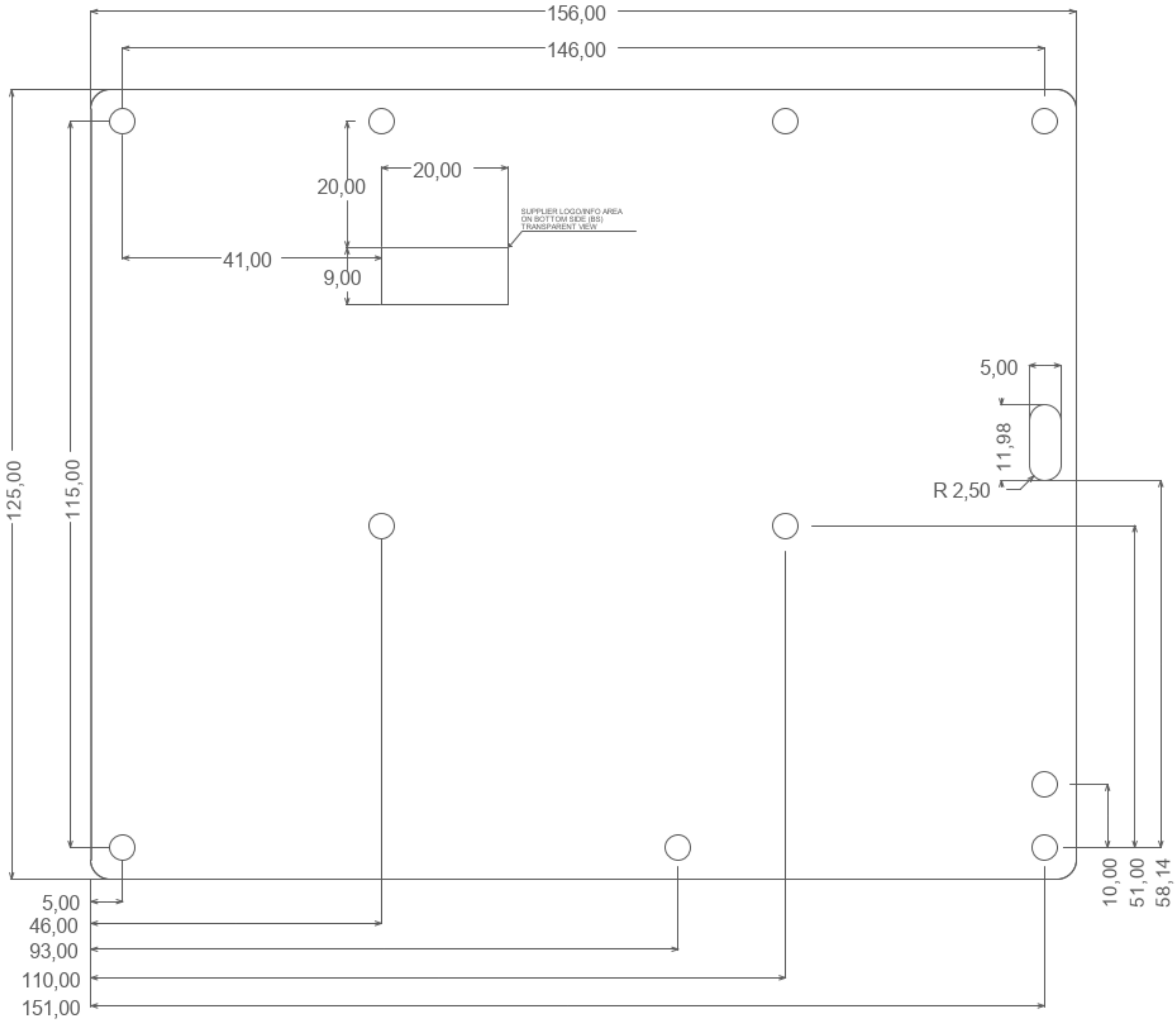


Figure 39: Board Mechanical Drawing (dimensions are in millimeters)

11 Acronyms and Abbreviations

Table 12: Acronyms and Abbreviations

Acronym	Definition
EVB	Evaluation Board
FTDI	Future Technology Devices International
I/O	Input Output
SIM	Subscriber Identification Module
TLB	Translation Board
UART	Universal Asynchronous Receiver Transmitter
USB	Universal Serial Bus

12 Related Documents

Refer to https://dz.telit.com/ for current documentation and downloads.

Table 13: Related Documents

S.no	Book Code	Document Title
1	1VV123456	Document Title 1
2	1VV123456	Document Title 2
3	1VV123456	Document Title 3

13 Document History

Table 14: Document History

Revision	Date	Changes
4	2023-11-10	Template upgrade
3	2022-11-16	Note added on B2B connector pinout table and USB connector on TLB
2	2022-10-04	Added a note related to the use of GPIO1/2 for audio codec Added a note related to the effect, at the module startup, of the capacitor connected to the push buttons
1	2022-06-23	Added a note related to the SIM detection
0	2022-03-16	First document revision

From Mod.0818 Rev.11